3. Skills, Connectors, and Plugins

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Codified Expertise, Not a Smarter Model

The problem with chatting

- A great prompt works once, then scrolls out of your history
- The next person — or the next you — starts over
- Quality depends on remembering how you did it last time

What a skill fixes

- Write down “how a good referee report is structured,” once
- Every future run follows it — same standard, no reminding
- A **one-time workflow** becomes a **reusable asset**

A skill is not a model upgrade. It is your procedure, written down where Claude will follow it every time — across Claude.ai, Claude Code, and the add-ins.

What a Skill Actually Is

A folder and a file

- A `SKILL.md` file: a name, a description, and instructions in plain English
- Optional scripts and reference documents alongside
- That is the whole thing — readable, editable, versionable

How Claude uses it

- At startup it reads only each skill's name and description
- When a task looks relevant, it pulls in the full instructions
- Scripts run only when needed — so your context stays lean

This "progressive disclosure" is why you can have dozens of skills installed and pay attention-cost only for the one in use.

Public Repos

Researchers Have Already Built These

Pedro Sant'Anna

- Runs his whole paper lifecycle through Claude Code
- `/lit-review`, `/review-paper` (mimics AER/QJE/JPE standards), referee responses
- ~18 specialized agents in an adversarial critic-fixer loop; blocks anything scoring below 80

Lee Crawfurd

- Five tightly-scoped skills for development economists
- Paper Review, Code Review (Stata/R/Python replication audit)
- "Referee 2" runs parallel reproducibility checks

Academic-research suite

- Deep Research with PRISMA systematic-review mode
- Drafting with citation-format conversion (APA/Chicago/MLA)
- A mock editor + reviewer + devil's-advocate panel

These are real, published, installable suites — economists mostly. The shared theme: Claude is **copilot, not pilot**. You own the question and the identification strategy; the skill handles the grunt work and enforces quality gates.

The Feature That Matters Most for Us

The biggest academic risk with AI is **fabricated references** and claims a cited source does not actually support. The better skills attack this head-on.

Verify every citation

Checked against Semantic Scholar, OpenAlex, Crossref, and arXiv — bogus DOIs caught before export

Audit every claim

Does the cited source actually support the sentence? High-severity violations block the output

Flag contamination

Detects citations that may themselves be AI-invented

For empirical or b-school research, this is the line between a demo and something you can responsibly use.

Does It Actually Work?

The benchmark

- 54 social-science papers, tested for whether a coding agent could reproduce the published findings from the paper alone
- Claude Code: **~93%** accuracy at the individual-step level
- But only **~78%** at the whole-paper level

What the gap teaches

- It nails each analysis step
- One broken step anywhere still fails the full reproduction
- A ready-made lesson on exactly where human oversight stays essential

The 93/78 split is the honest picture: extraordinary at the parts, not yet trustworthy end-to-end without a human checking the chain.

Build Your Own

From Workflow to Skill

The easy path

- Type `/skill-creator` and describe your workflow
- It writes the `SKILL.md` and saves it where Claude will find it
- Invoke it by name from then on

Bake in the checks

- Put your verification steps *inside* the skill
- “After computing, show the row count and the min and max”
- The skill enforces your hygiene, every run — not just when you remember

The most valuable skills for research encode not just what to do, but how to *check* it — the referee’s discipline, made automatic.

A Skill You Might Write

SKILL.md — replication-audit

```
When given a replication package: 1) inventory every script and data file; 2) run the code end
to end in a clean environment and record what breaks; 3) check each table and figure in the
paper against what the code produces; 4) flag any result that cannot be reproduced from the
provided files; 5) write a short report with a per-result reproduced / not-reproduced status.
```

A hand-written skill like this — yours, transparent, under your control — is safer and more teachable than installing a multi-skill black box.

Vetting

Installing Someone Else’s Skill Runs Their Code

The supply-chain risk

- The academic suites are independent repos, outside Anthropic’s screened directory
- Installing one runs someone else’s instructions and scripts on your machine
- There is active security research on malicious third-party skills

Due diligence

- `claude plugin validate` checks *structure*, not safety
- Read the `SKILL.md` and any scripts yourself
- Fine for a class demo; think twice before pointing it at real data

“Validate” confirms a plugin is well-formed, not that it is trustworthy. The safest research skill is often the small one you wrote and can read in full.

The Payoff

A skill turns your best research practice — how you referee, how you audit a replication, how you check a citation — into something **Claude does the same way every time.** You still own the question and the judgment. The skill owns the discipline.

Breakout

Questions to Discuss

- What is one research task you do the same careful way each time — a good candidate for a skill?
- Where has a fabricated or mis-supported citation nearly slipped past you? Would a claim-audit have caught it?
- Given the 93/78 result, which parts of your empirical work would you let AI run — and which would you always check yourself?
- Would you install a colleague's research skill? What would you read before trusting it with your data?